智能结构与智慧港口

Smart Marine Metastructures

浙江大学海洋学院
Ocean college, Zhejiang University

2025/2/25

CALENDERS

Week	Subject	Content
1	Introduction and Overview	1. Overview of structural and material analysis 2. Introduction of structural materials and vision for their applications in ocean engineering
2	Advanced Metastructures	1. Structures vs. metastructures 2. Advanced characteristics of metastructures
3	Structural Health Monitoring (SHM)	1. Advent and development of structural health monitoring (SHM) 2. Main SHM technologies
4	SHM in Civil and Ocean Engineering	1. SHM in civil and ocean engineering 2. Challenges and future work
5	Introduction to Intelligent Ports	1. Traditional ports vs. modern intelligent ports 2. Information networks of intelligent ports
6	Equipment and Cases of Intelligent Ports	1. Typical equipment in intelligent ports 2. Case study of intelligent ports
7	Structural Design and SHM in Data Era	1. Vision for smart marine metastructures and intelligent ports 2. Summary of the class
8	Oral Presentation	

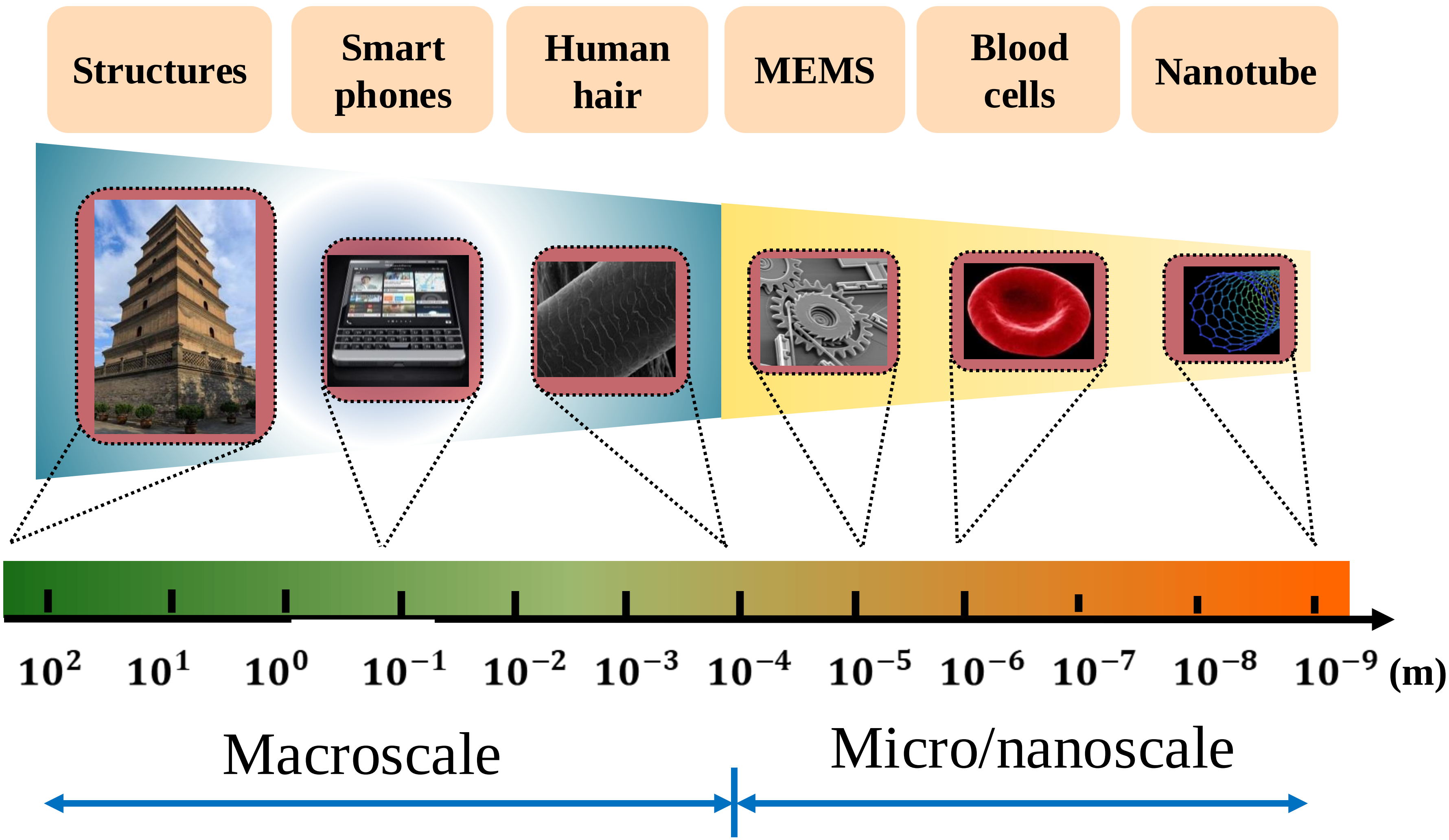
Smart Marine Metastructures

Metastructures

Marine

Smart

Our World


Structures
Smart phones
Human hair
MEMS
Blood cells
Nanotube
10^{2} 10^{1} 10^{0} 10^{-1} 10^{-2} 10^{-3} 10^{-4} 10^{-5} 10^{-6} 10^{-7} 10^{-8} 10^{-9} (m)
Macroscale
Micro/nanoscale

Structures

Demands

Port index decomposition

Structural planning and layout

Port index decomposition

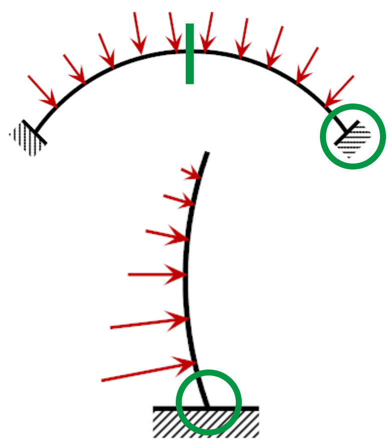
Marine engineering building

Architecture index decomposition

Material, structure

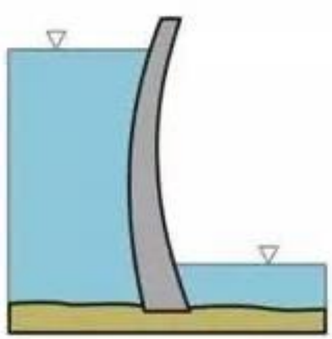
Structural mechanics

Distribution of stress and deformation



Service condition

Given loading



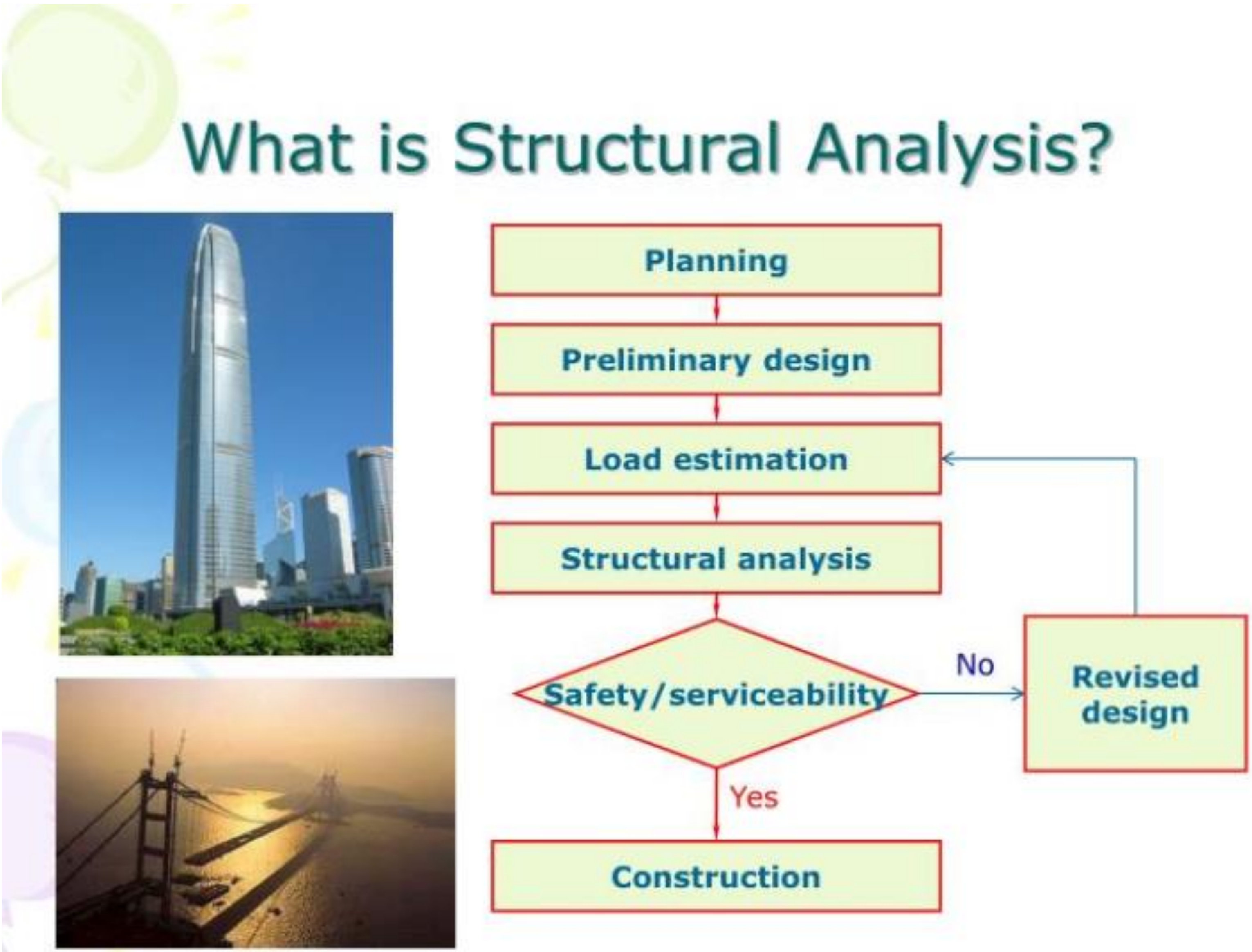
Reinforced concrete

Load bearing material design

Rock and Soil Mechanics

Foundation support analysis

Structures

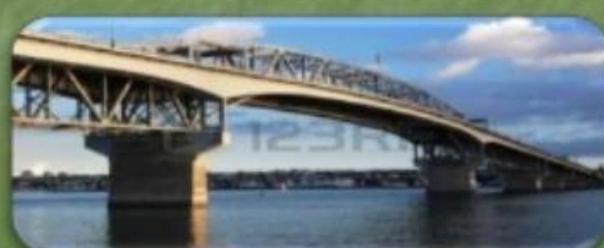

What is Structural Analysis?
Planning
Preliminary design
Load estimation
Structural analysis
Safety/serviceability
No
Revised design
Yes
Construction

Structures and Analysis

#1 - INTRODUCTION TO STRUCTURAL ANALYSIS

Structural Analysis is the analysis of a given structure subject to some given loads, and the idea is to predict the response of the structure (system), thus there are some inputs referred to as ''*stimulus*'' and an outputs referred to as ''*response*''. The Structural analysis is the application of solid mechanics to predict the response (Interms of forces & displacements) of a given structure (existing or proposed) that subjected to a specified loads.

LOADS
(INPUTS)



RESPONSE
(INPUTS)

STRUCTURE (SYSTEM)

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Structures and Analysis

Loads are classified into 2 parts:

- Dead loads (Self-weight; of the components of the structure).
- Imposed loads & forces (Live, Wind, Snow, Rain, and Temperature, Erection loads, Seismic forces & others).

The Structural Analysis is only ascertained correct when the following requirements were satisfied;-

- Equilibrium Forces.
- Compatibility of Displacement.
- Force/Displacement Relations.

Therefore, In every forms of the structures, the structural Engineer has to consider the following structural considerations and purposes.

Structures and Analysis

- Stability,
- Strengths,
- Stiffness,
- Economy, as well as
- Aesthetic aspects of the structure.

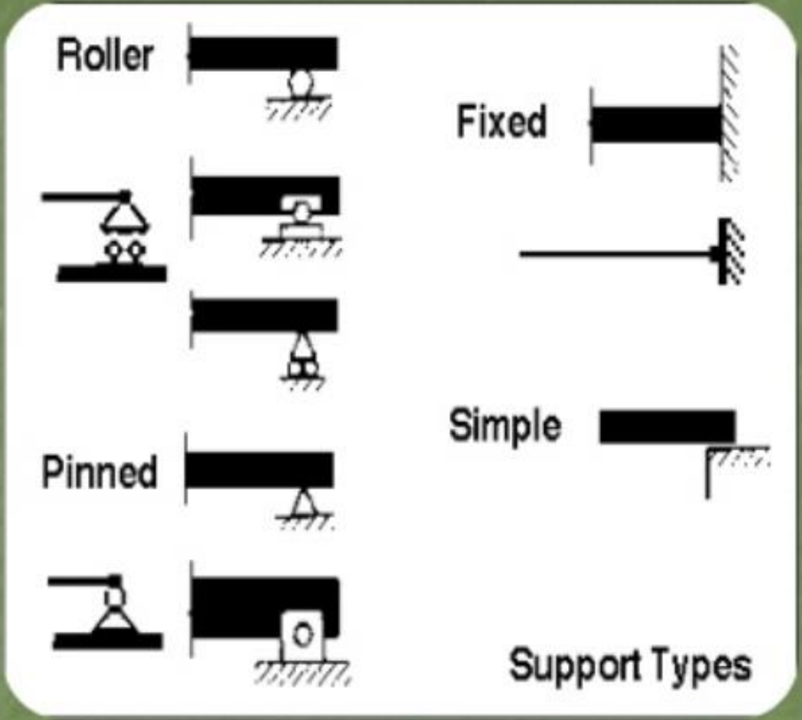
The following Softwares are available

- Orion,
- STAAD III,
- ANSYS,
- AXIS VM,
- ETABS, SAP, Etc

Structures and Analysis

#2 – DIFFERENT TYPES OF SUPPORTS

- Roller,
- Pinned,
- Hinged,
- Fixed,
- Link,
- Ball and Socket,
- Rigid Support,
- Spring Support.

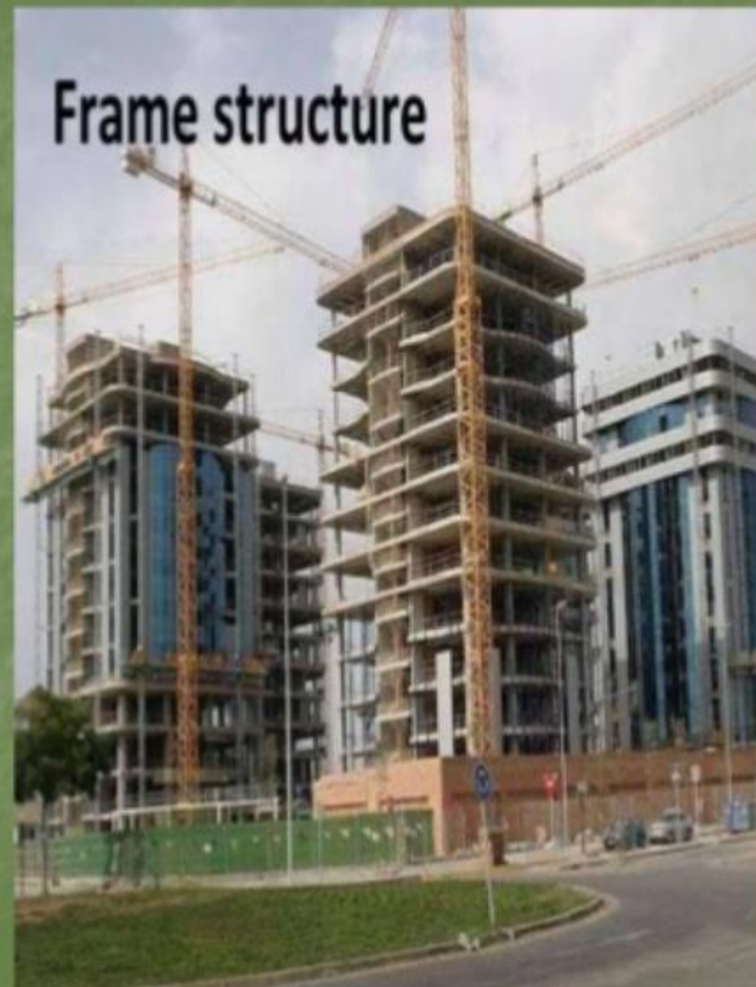


FORCES

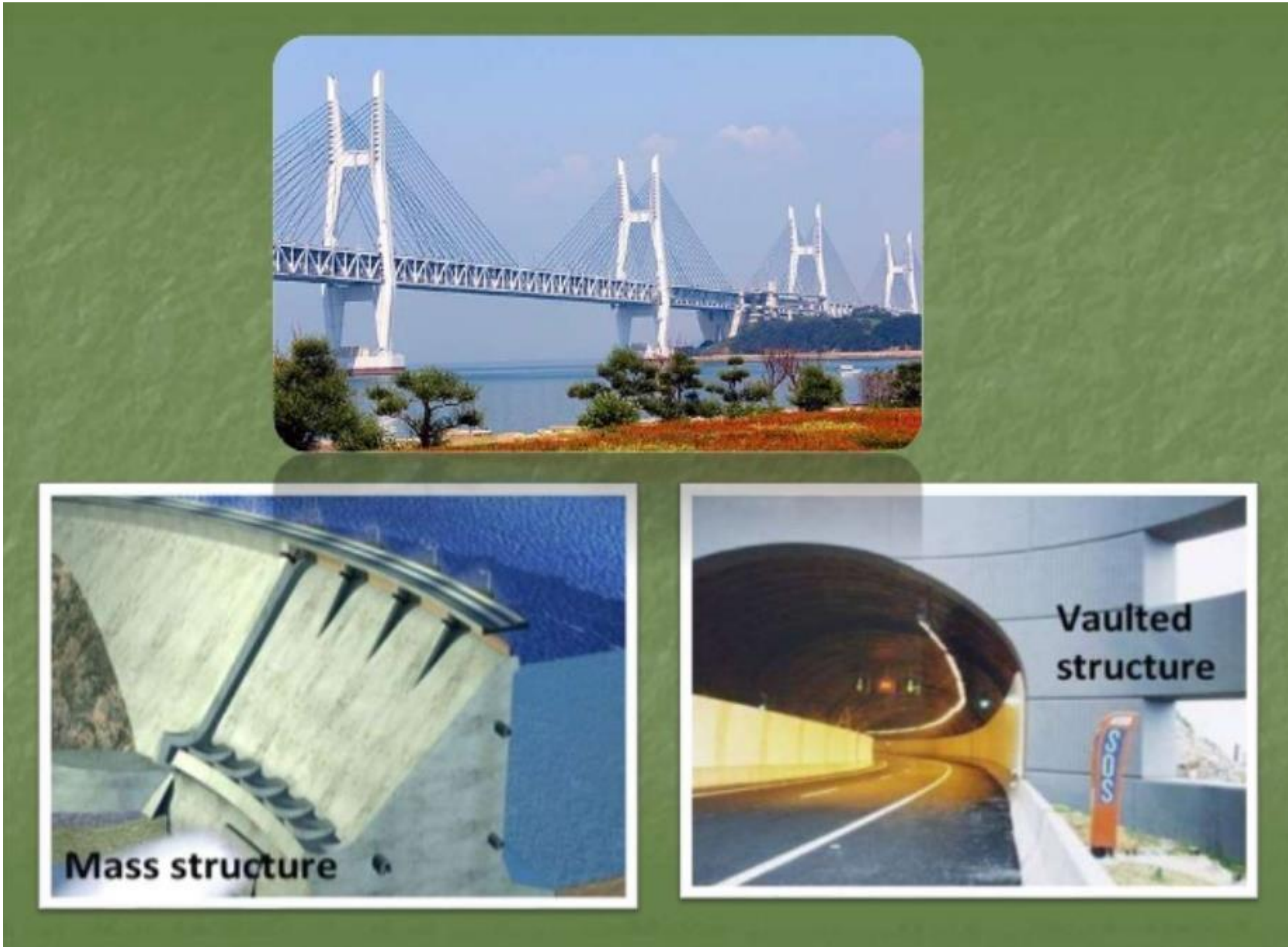
Structures and Analysis

#2 – DIFFERENT TYPES OF STRUCTURES

- Frame Structure,
- Truss Structure,
- Shell Structure,
- Arch Structure,
- Suspension Structure,
- Mass Structure, &
- Composite Structure.



Structures and Analysis

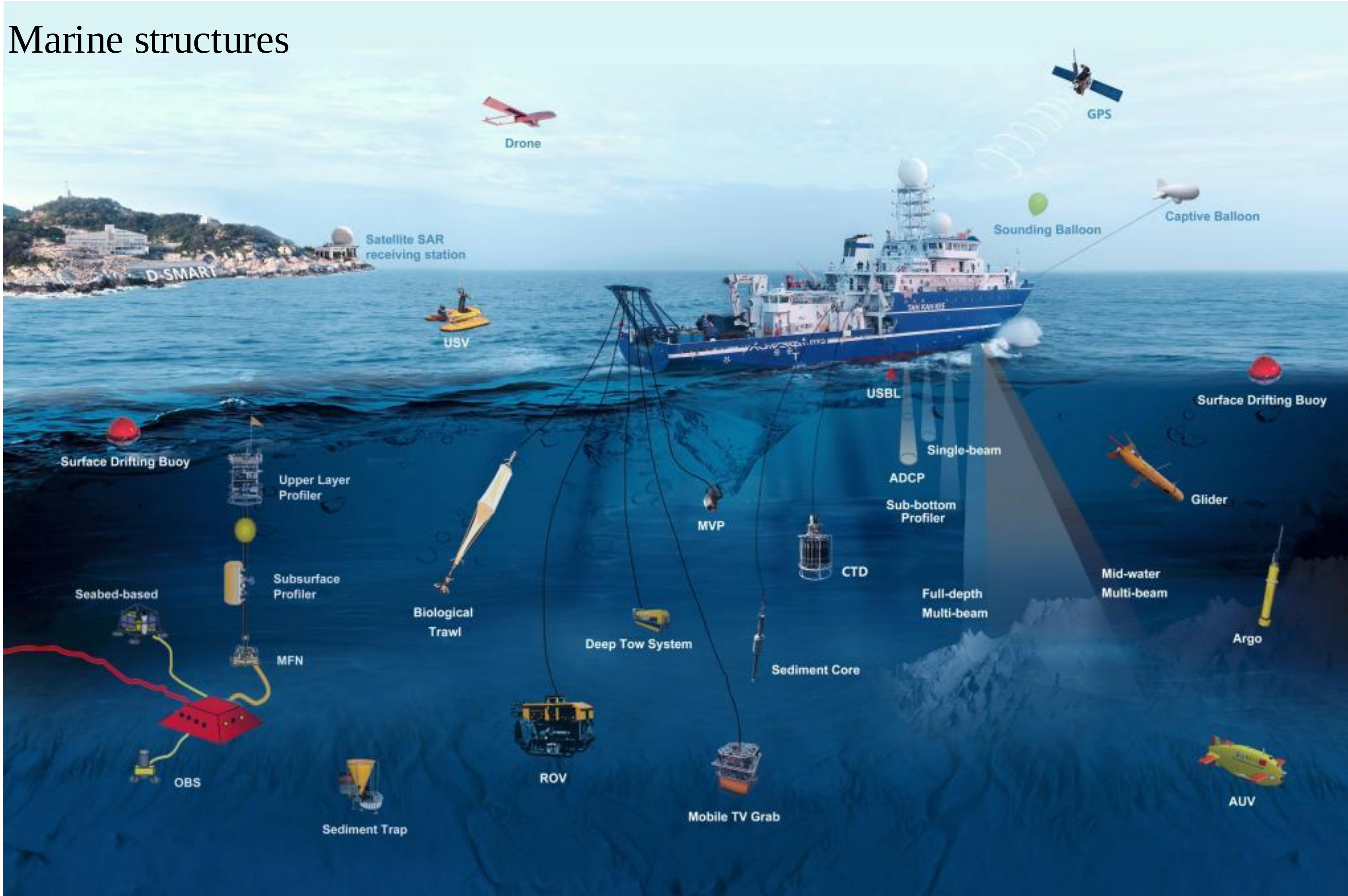


Structures and Analysis

Marine Structures

Structures and Analysis

Marine structures



Structures and Analysis

Marine environment protect

Structures and Analysis

#3 – LIST OF EQUILIBRIUM EQUATIONS

For state of static equilibrium to exist, it is necessary that the combined resultant effect of the system of forces shall be neither a **Force** nor a **Couple**; otherwise there will be tendency for motion of a body. However, many 3D structures are idealized as series of 2D components:

- If the combined resultant effect of a general system of forces acting on a planar structures is not be equivalent to a resultant force, the algebraic sum of all F_x components must be equal to Zero and the algebraic sum of all F_y components must to be equal to Zero, i.e $\sum F_x = 0 \quad \sum F_y = 0$.
- If the combined resultant effect is not to be equivalent to a couple, the algebraic sum of the moment M_z of all forces about any axis parallel to the Z-reference axis and normal to the plane of the structure must be equal to zero, i.e $\sum M_z = 0$.

Structures and Analysis

#4 – DETERMINATE AND INDETERMINATE OF A STRUCTURE

If the number of unknown forces is identical to the number of equilibrium equations, the problem is said to be ***Statically Determinate*** since the unknown forces can be determined directly from the equilibrium.

$$2j = m + r \qquad where\ n = 0$$

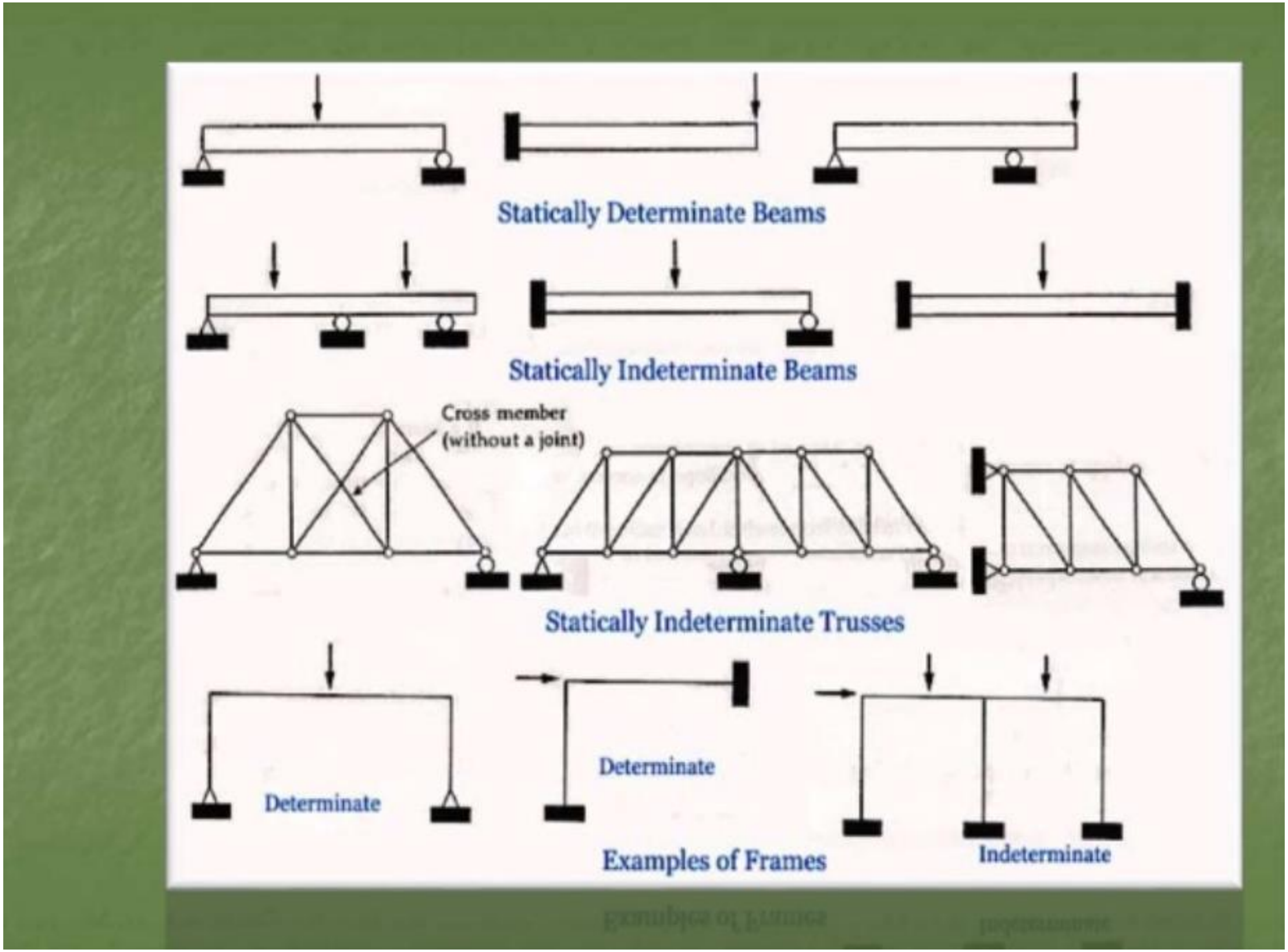
If, however the number of unknown forces exceeds the number of equilibrium equations, the problem is said to be ***Statically Indeterminate*** to a degree equal to this excess.

$$2j > m + r \qquad where\ n \neq 0$$

$$j = No.of\ joints, \qquad m = No.of\ members,$$

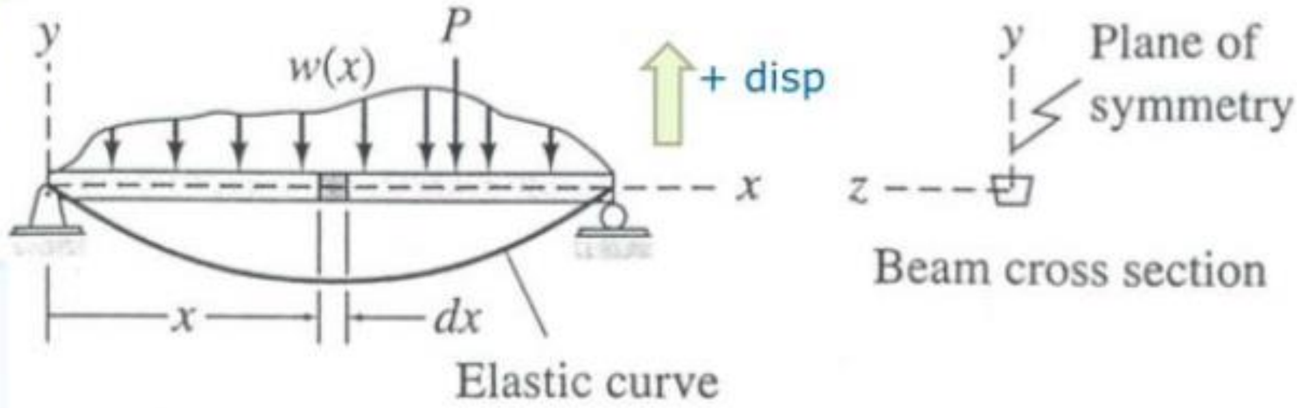
$$r = No.of\ reactions$$

Structures and Analysis



Structures and Analysis

Direct Integration

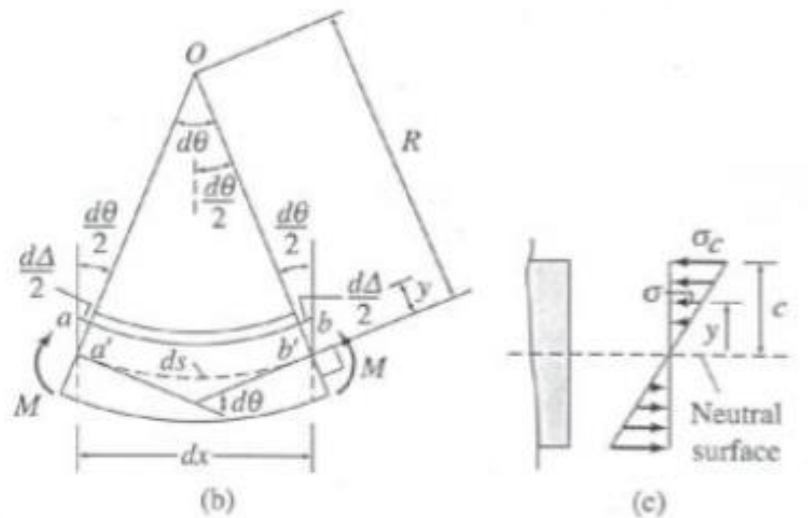


Bernoulli-Euler beam

$$\frac{d^2y}{dx^2}=\frac{M}{EI} \qquad \frac{d\theta}{dx}=\frac{M}{EI}$$

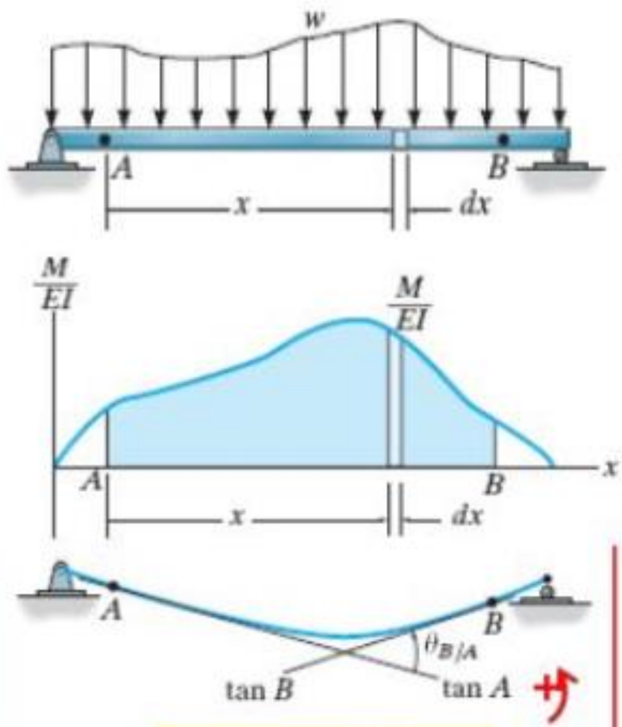
$$y=\iint\frac{M}{EI}dx\,dx \qquad \theta=\int\frac{M}{EI}dx$$

apply boundary conditions



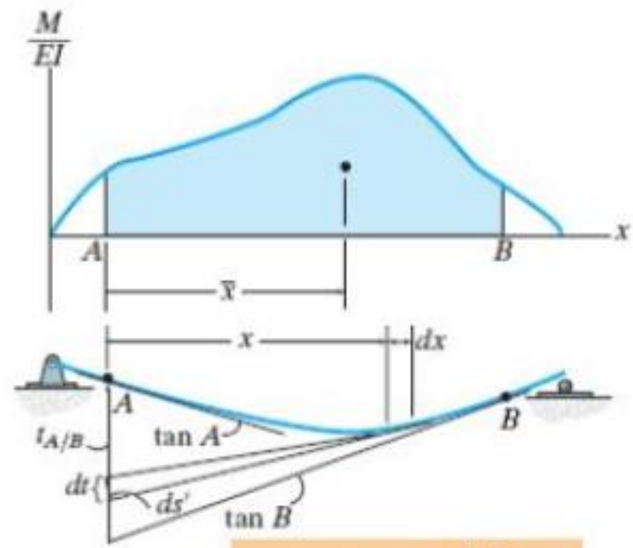
Moment-Area Methods

1st moment-area theorem



$$\theta_{B/A} = \int_A^B \frac{M}{EI} dx$$

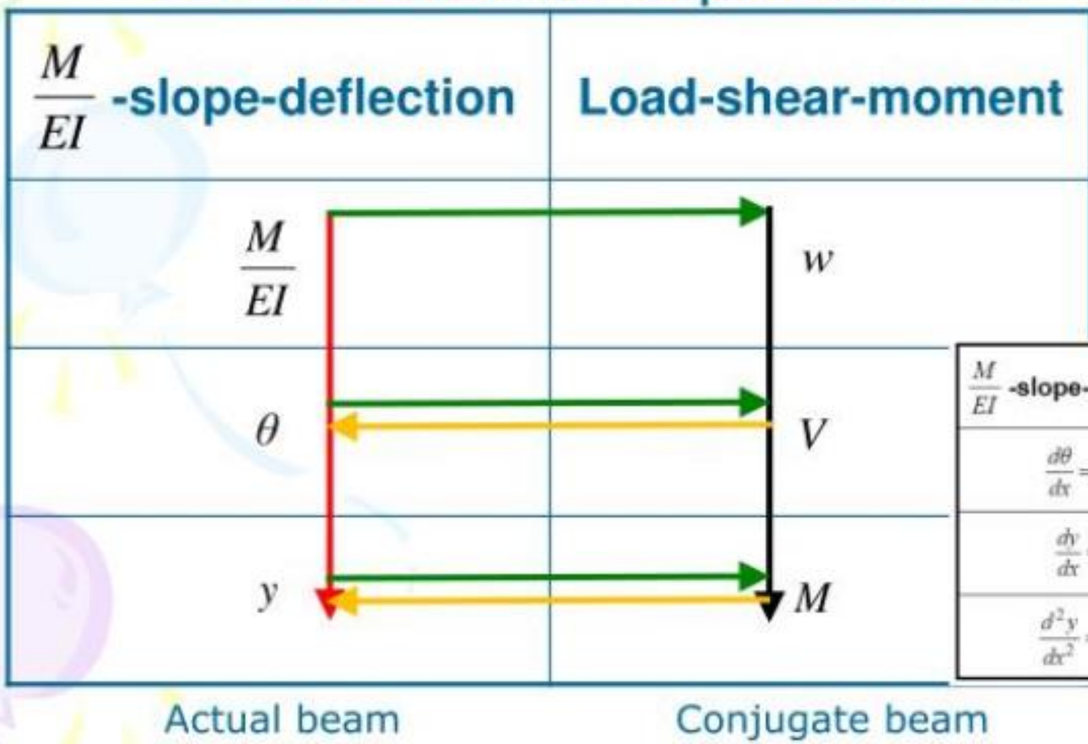
2nd moment-area theorem



$$t_{A/B} = \bar{x}\int_A^B \frac{M}{EI} dx$$

Conjugate-Beam Method

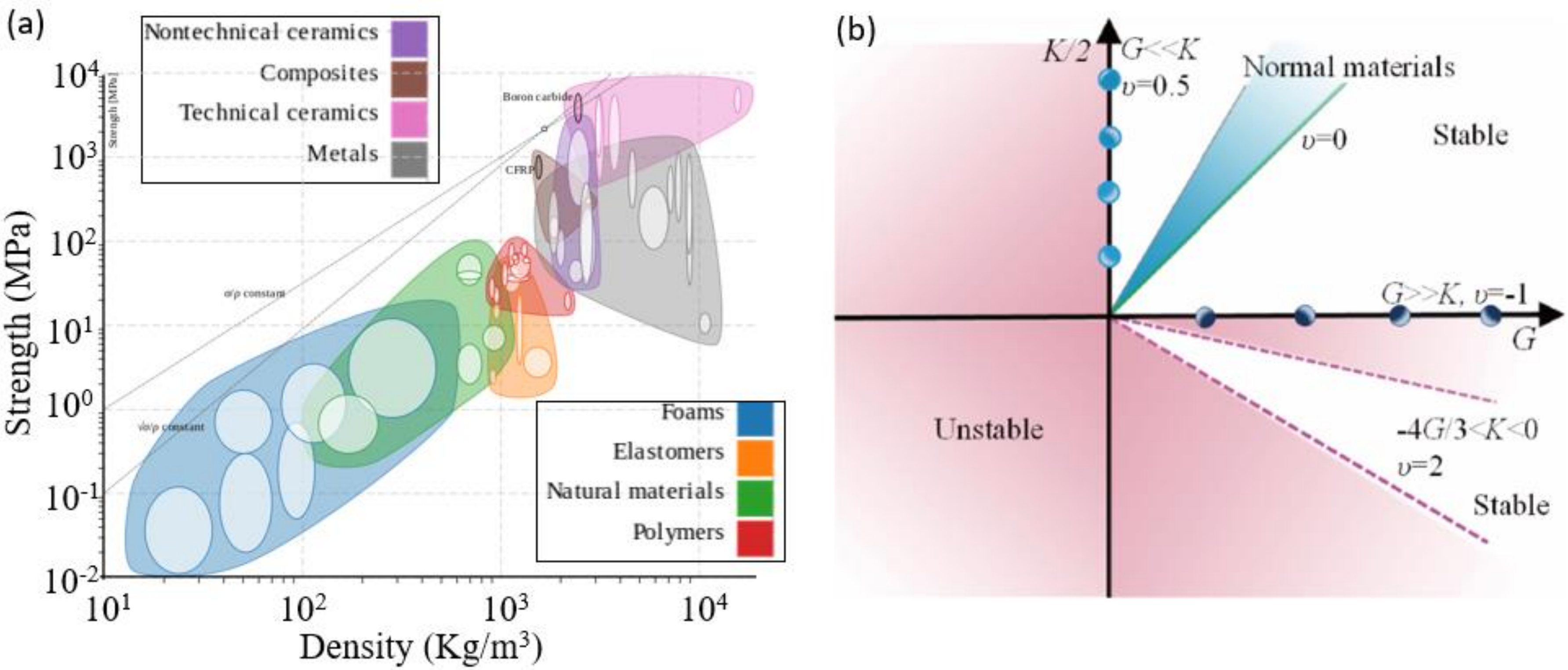
- Mathematical equivalence



$\frac{M}{EI}$ -slope-deflection	Load-shear-moment
$\frac{d\theta}{dx}=\frac{M}{EI}$	$\frac{dV}{dx}=w$
$\frac{dy}{dx}=\theta$	$\frac{dM}{dx}=V$
$\frac{d^2y}{dx^2}=\frac{M}{EI}$	$\frac{d^2M}{dx^2}=w$

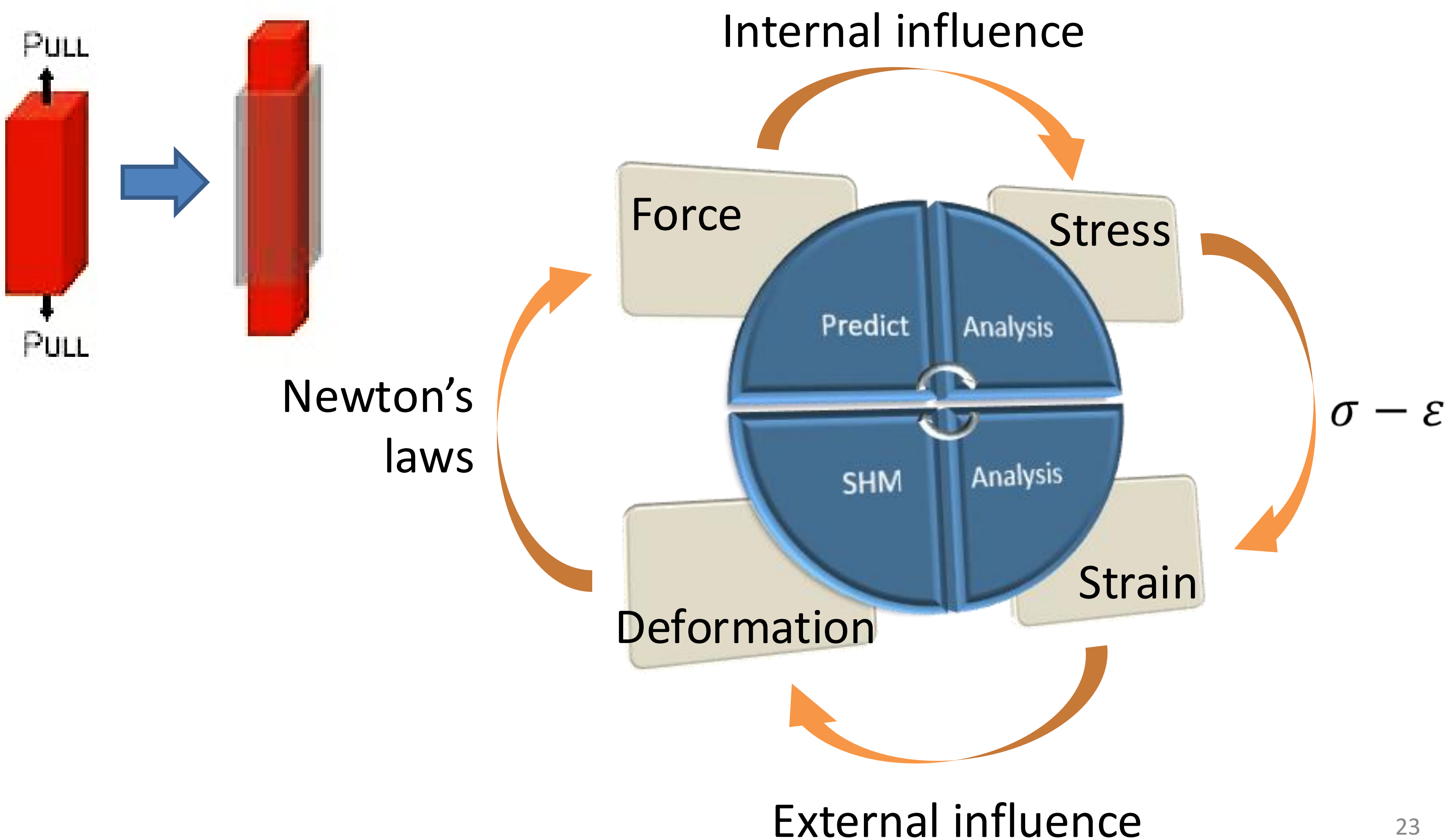
Materials and Analysis

Background



Materials and Analysis

Background

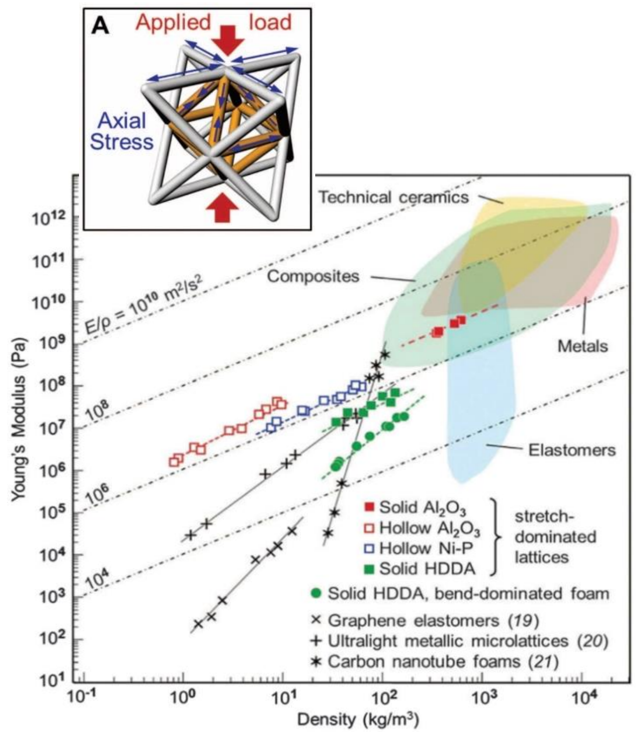


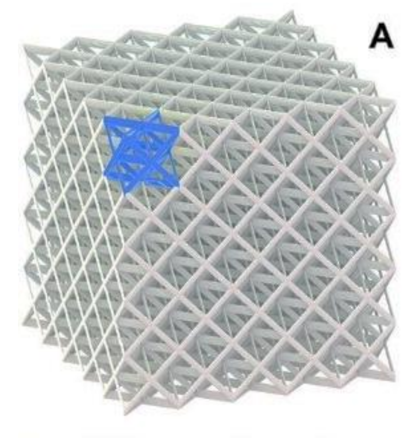
Materials and Analysis

Materials

- Composites
- Functional graded materials
- Graphene, Nanotube
- et al.

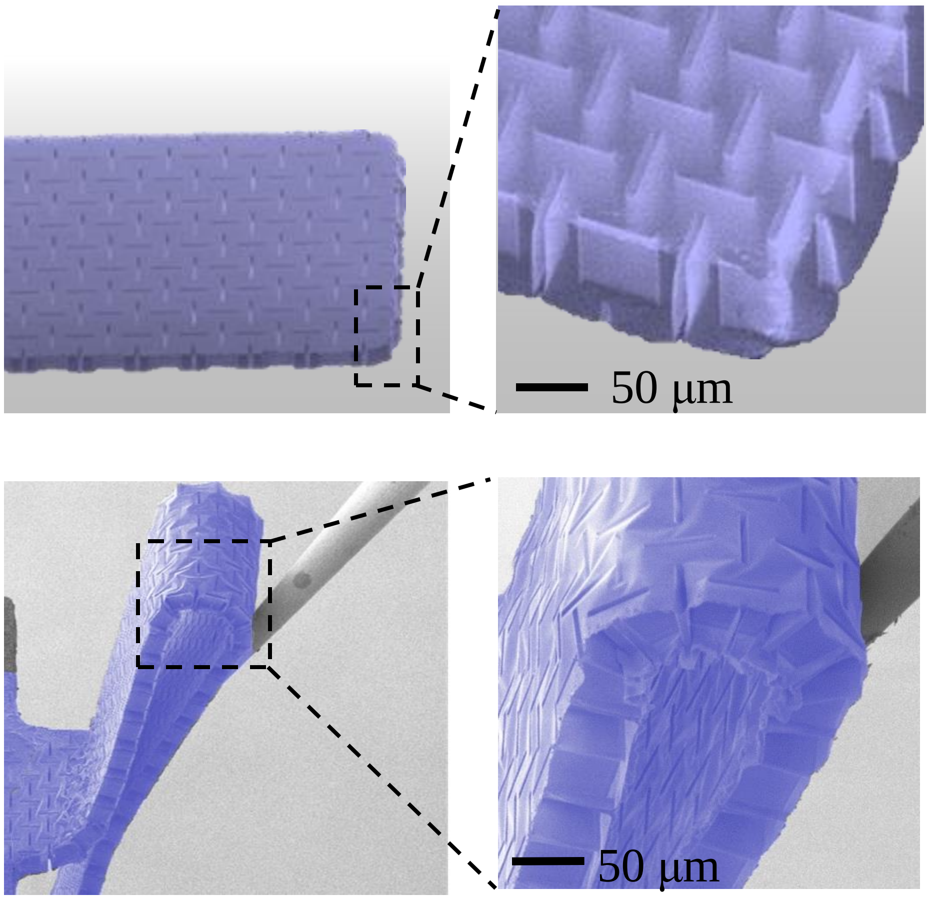
Structures

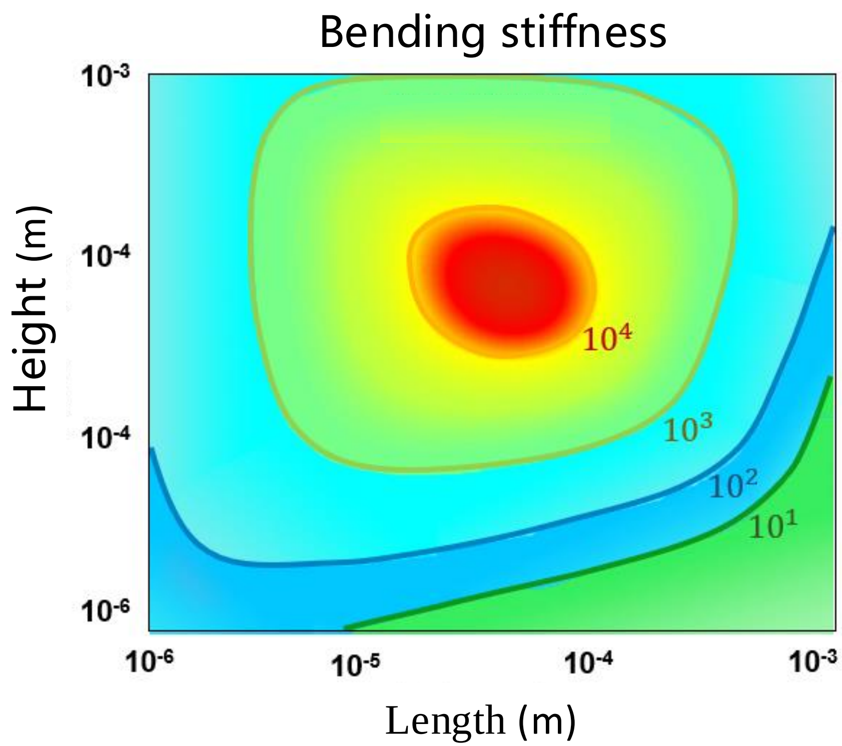




Zheng et al. (2014). *Science* 344(6190): 1373-1377.

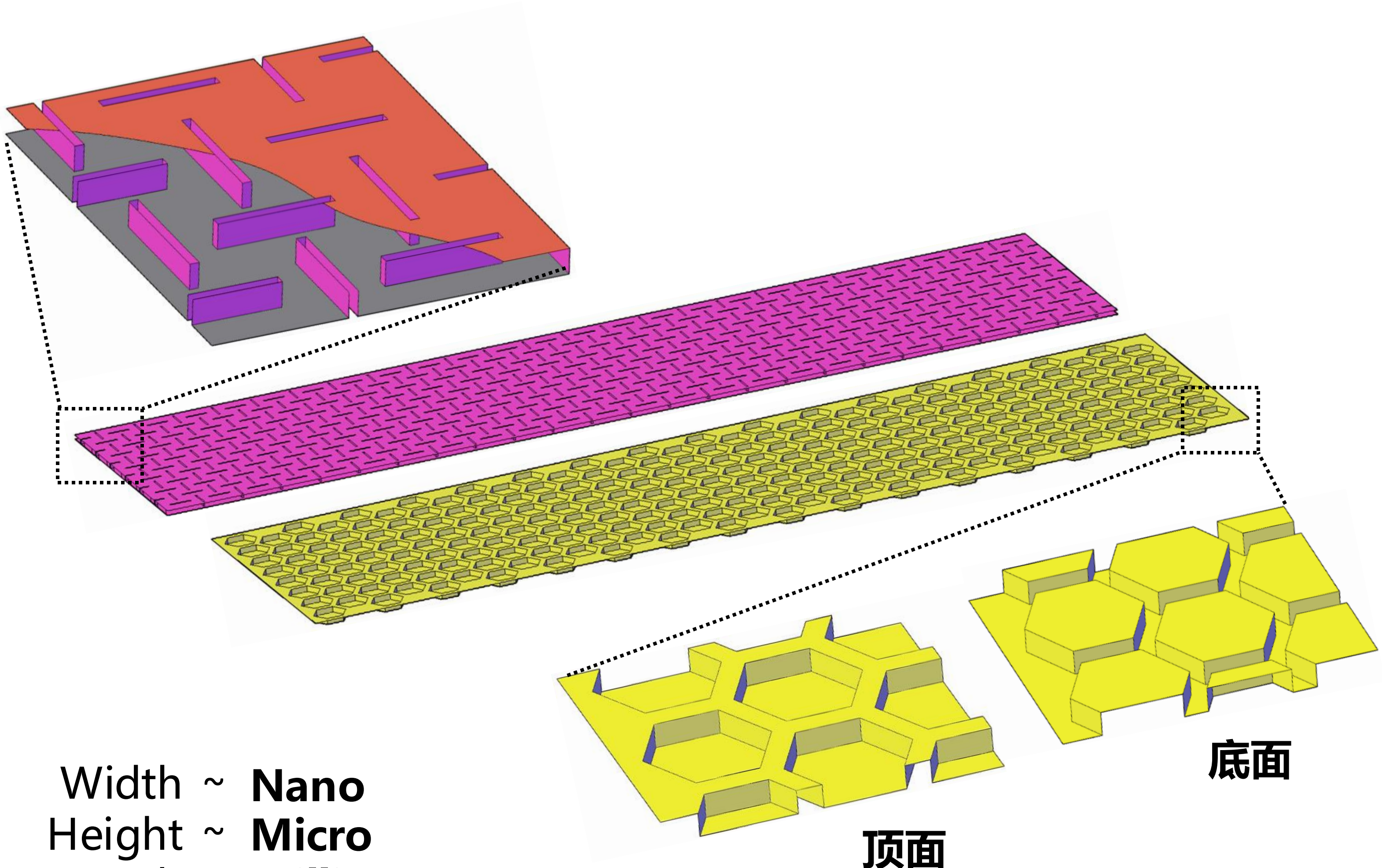
Materials and Analysis





100-nm-thick Alumina 3-µm-tall
Nanocardboard Cantilever
Video Accelerated 8x

Materials and Analysis



Width ~ **Nano**
Height ~ **Micro**
Length ~ **Millimeter**

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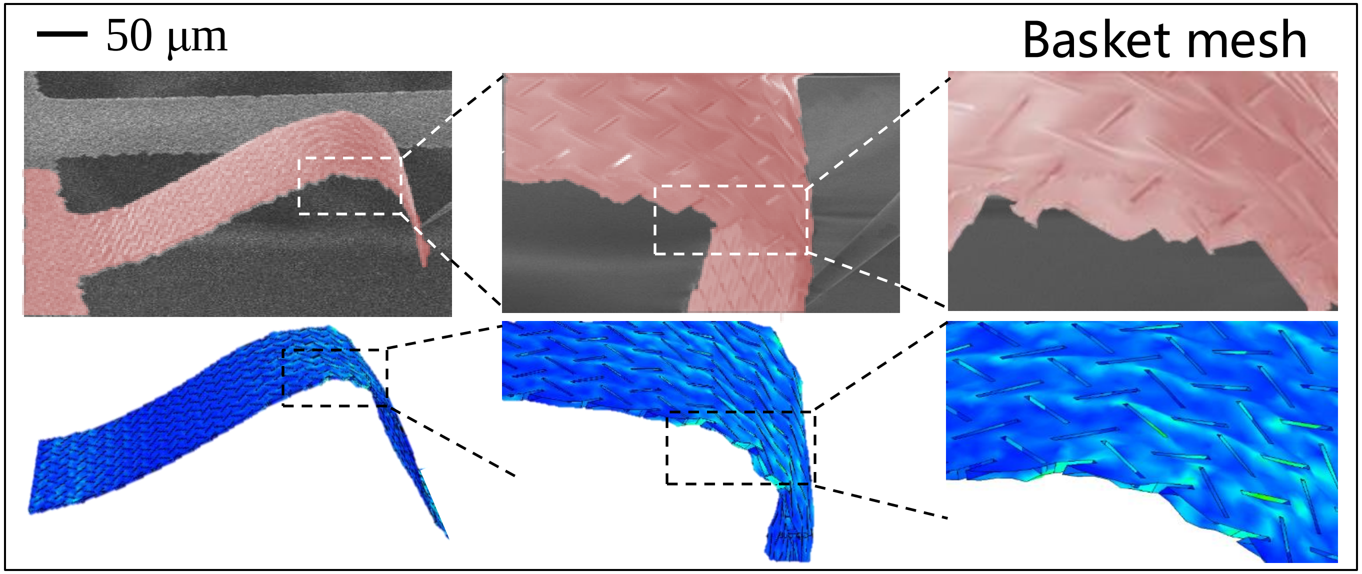
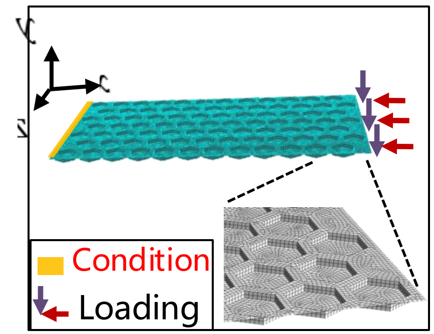
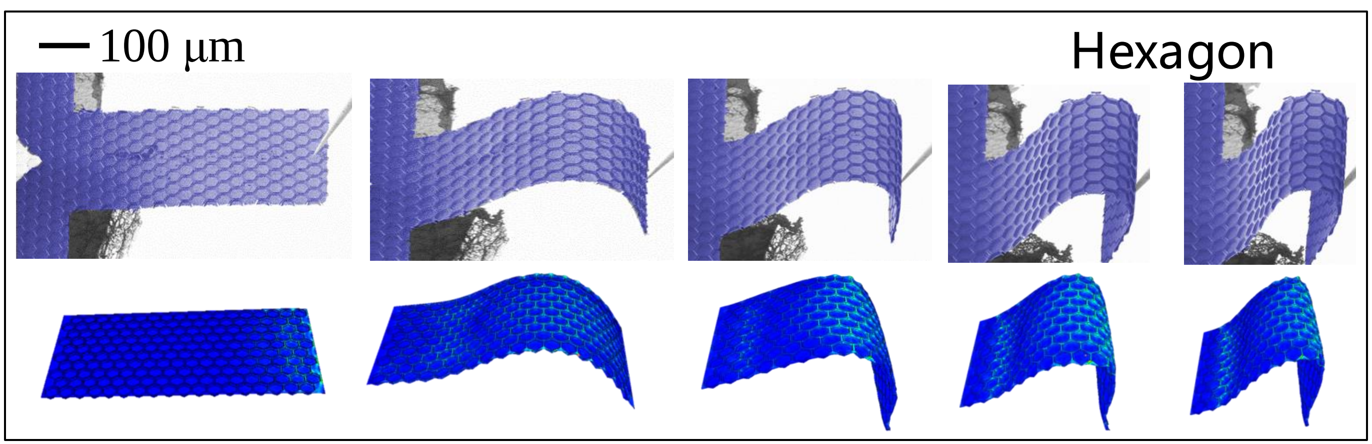
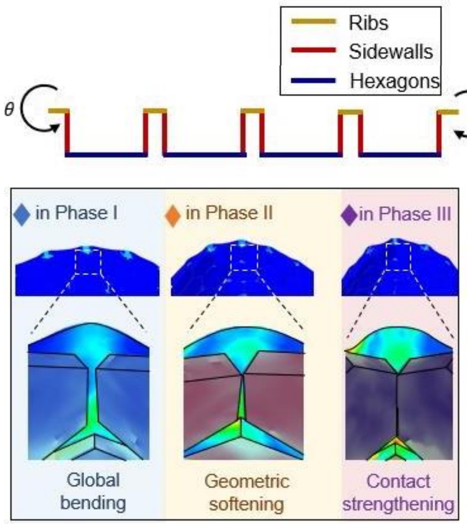
Materials and Analysis

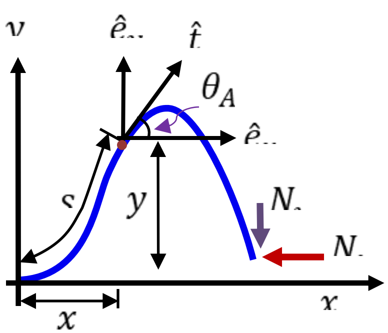




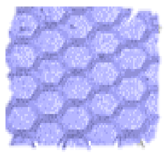


Materials and Analysis





Hexagon



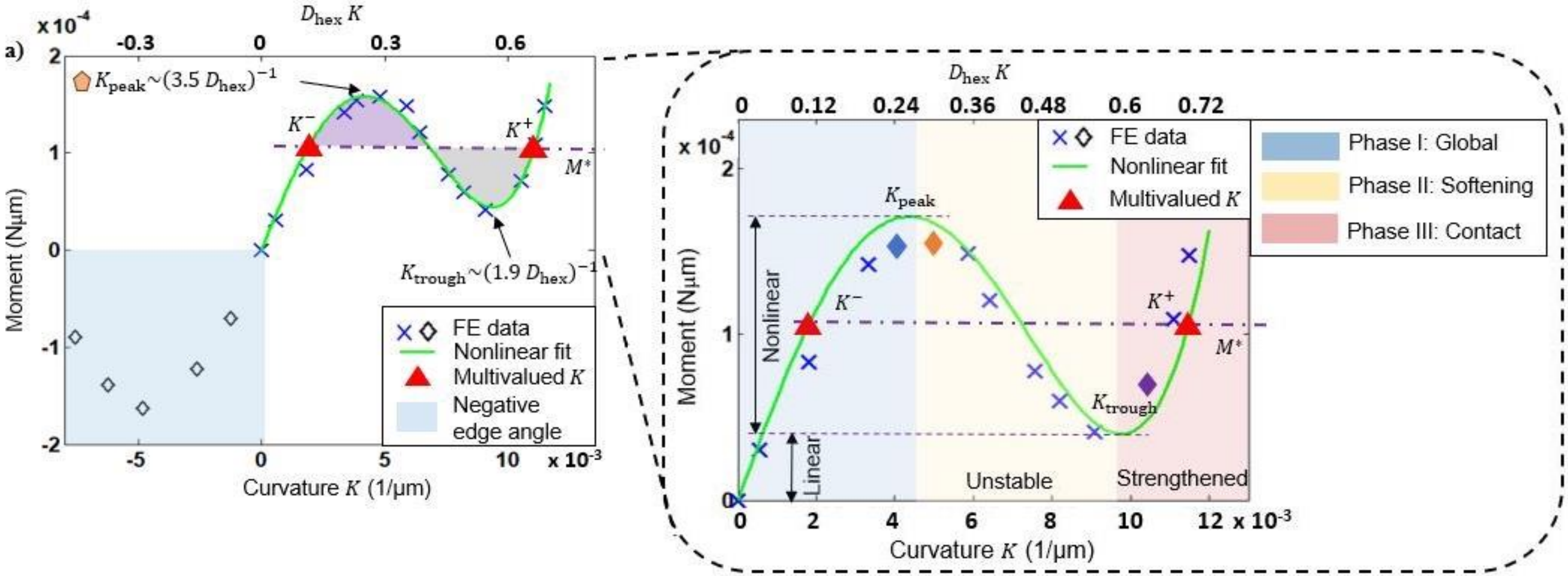
$$D\frac{\mathrm{d}^2\theta}{\mathrm{d}s^2}\left[1-0.74D_{\mathrm{hex}}\frac{\mathrm{d}\theta}{\mathrm{d}s}-21.69{D_{\mathrm{hex}}}^2\left(\frac{\mathrm{d}\theta}{\mathrm{d}s}\right)^2+35.28{D_{\mathrm{hex}}}^3\left(\frac{\mathrm{d}\theta}{\mathrm{d}s}\right)^3\right]+\cos(\theta)N_y-\sin(\theta)\,N_x=0$$

$$\text{where } D=\frac{Et^3}{12(1-\nu^2)}\left(\frac{D_{\mathrm{hex}}}{W_{\mathrm{rib}}}+1\right)^2$$

$$\begin{cases} X_2 = X_1{}' \\ X_2{}' = \dfrac{\sin(X_1)\,N_x-\cos(X_1)N_y}{D\left(1-0.74D_{\mathrm{hex}}X_2-21.69{D_{\mathrm{hex}}}^2{X_2}^2+35.28{D_{\mathrm{hex}}}^3{X_2}^3\right)} \end{cases}$$

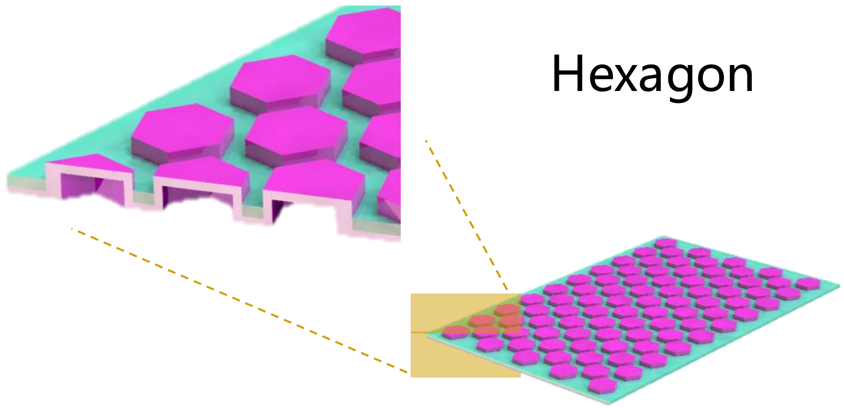
where $X_1=\theta(s)$, $X_2=\theta'(s)$ and $\nu=0$

Materials and Analysis

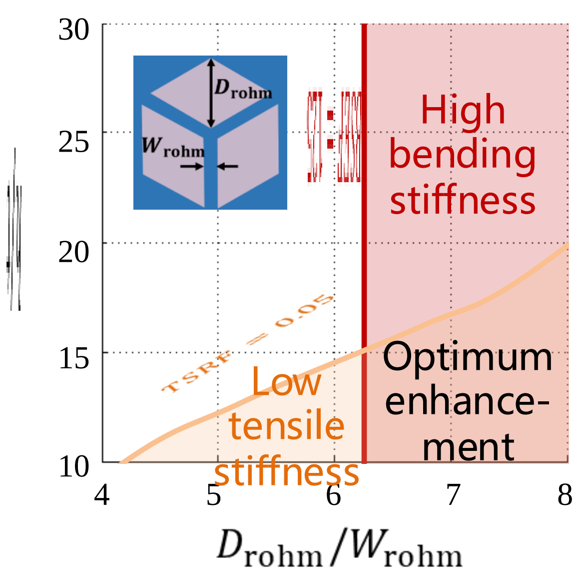


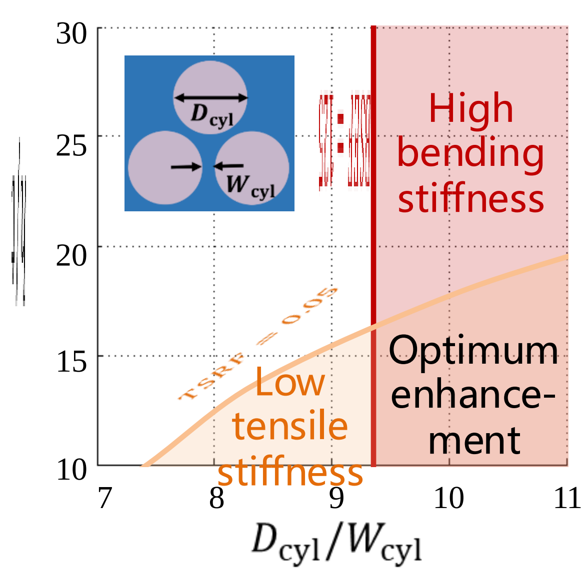
$$M = DK\left(1 - 0.37D_{\text{hex}}K - 7.23{D_{\text{hex}}}^2K^2 + 8.82{D_{\text{hex}}}^3K^3\right)$$

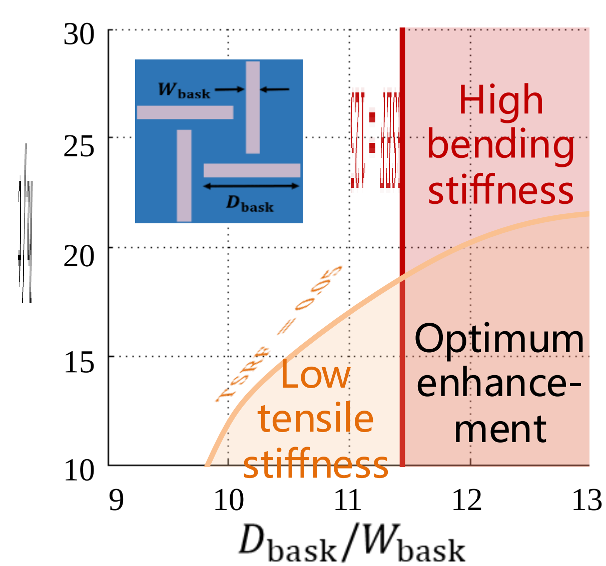
Materials and Analysis



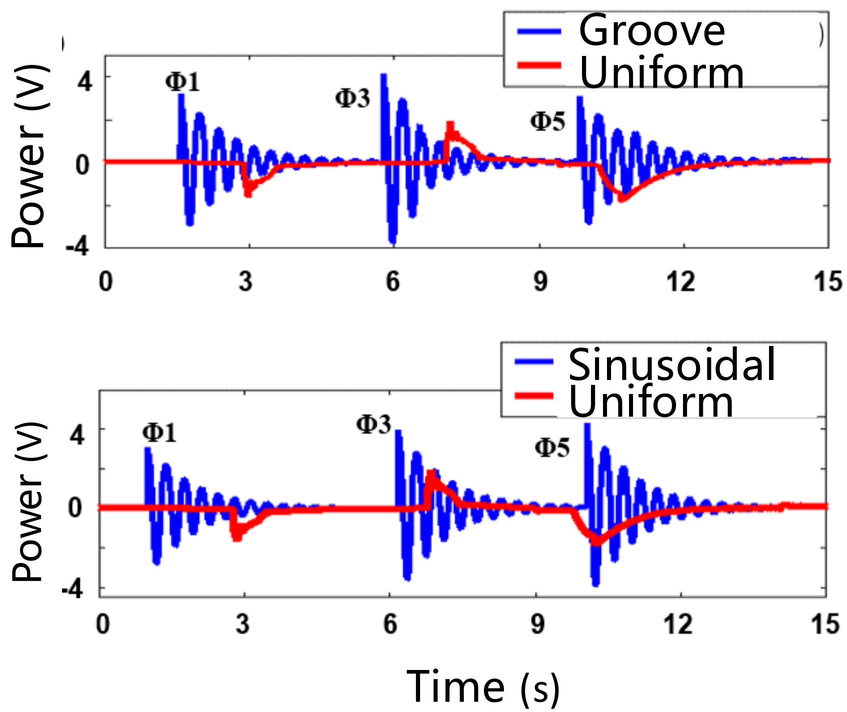
Hexagon

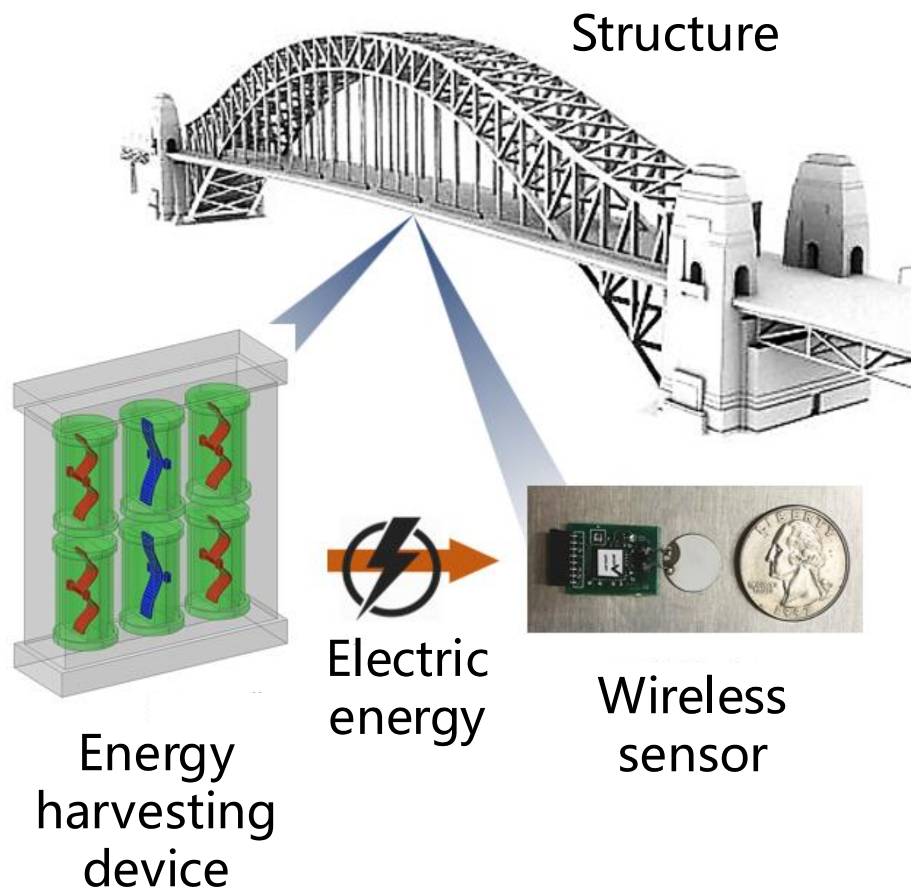






Applications





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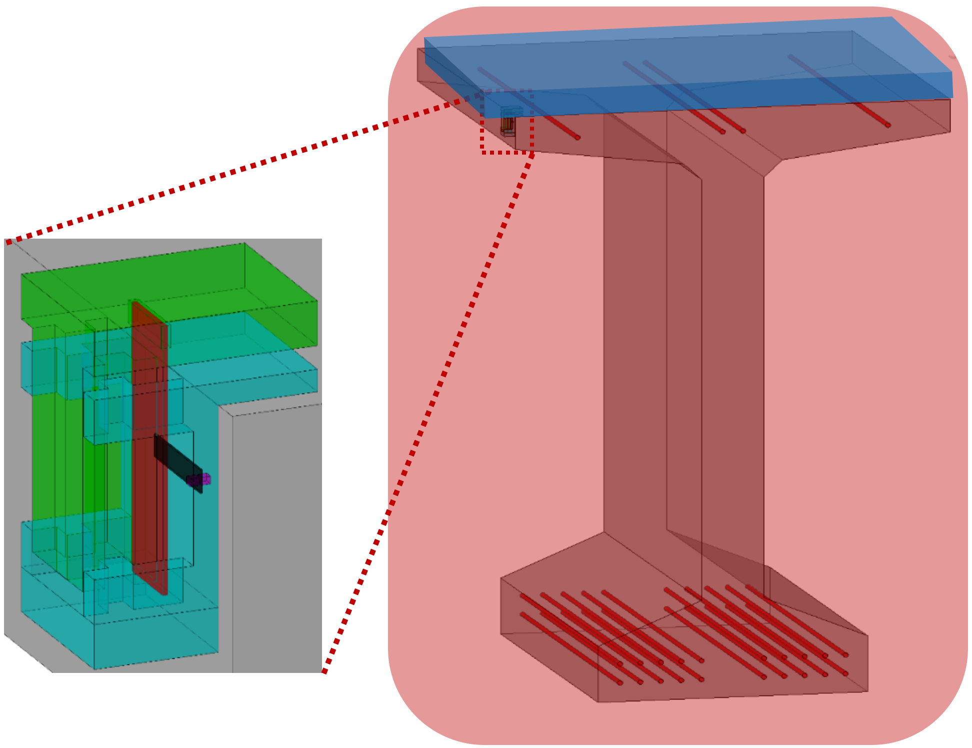
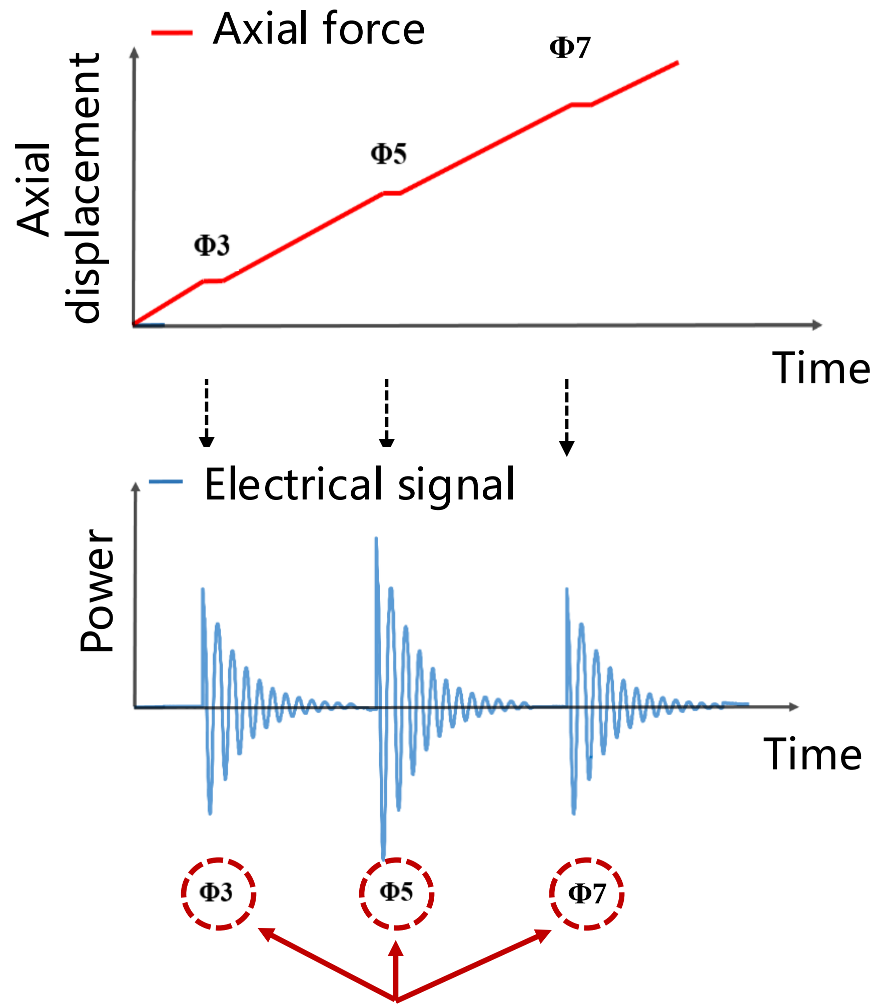
Applications



Thermal load



Thermal limit state

Applications

Smart Cities

END

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